

Änderungsbeschreibung/Description of Revision war kZ		Kommt vor/Frequency
Buchst./Rev.Ltr. -22	Änderungs-Nr./Revision Notice No. PR033847	Bearbeitungsstatus/Lifecycle Released

Corrections / New Items
First version of Genoline New Generation – Foreign MCS J1939 Interface

Allgemeintoleranzen/General Tolerances ISO 2768-mK				Masse/Mass	kg	Format/Size
Tolerierung/Tolerancing ISO 8015		Masse/Size ISO 14405		A4		
Ausf. u.Lieferung Tech. Characteristics		Oberflaechenangaben nach MTN5033 Surface Specification per MTN5033		Werkstoff Material		
Oberflaechenschutz Surface Protection				Halbzeug,Modell Gesenk Semifinished Product Pattern, Die		
Verwendbar f.Type Applicable to Model		Allgemeine Fertigungsvorschriften nach MMN332 Production Specification per MMN 332		Material-Nr./Material No.		
Projekt-/Auftrags-Nr. Project/Order No.				BZ00E52000011		
Ref.-Nr./ Ref. No.	Emission ID					
-						
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		Erstell. Drawn	2017-06-28 10:08:36	kellerma	/ Genoline NewGeneration - FOREIGN MCS J1939 - Signal- Mapping - Type1 Var01 ED01_00 GLNG FOREIGN MCS J1939	
		Bearb. Change	2018-09-18 08:26:47	schaeppers		
		Norm Std.	2018-09-18 08:26:47	ihmor stef		
		Gepr. Checked	2018-09-18 08:26:47	ihmor stef		
		Zeichnungs-Nr./Drawing No.			Blatt/ Sheet	
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		ZNG00007154				
		Type1 Var01 ED01_00				

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J1939 PGN Tx Signals

PGN-Name	PGN	PDUFormat	PDUspecific	DataPage	DefaultPriority	TransferMode	TxRepetitionRate	SendOnChangeTime	DLC
Engine Hours, Revolutions	65253	254	229	0	6	5	0	0	8
Exhaust Port Temperature 3	65185	254	161	0	7	1	1000	0	8
Dash Display	65276	254	252	0	6	1	1000	100	8
Engine Electronic Controller #1	61444	240	4	0	3	1	100	0	8
Engine Temperature 1	65262	254	238	0	6	1	1000	0	8
Fuel Information (Liquid)	65203	254	179	0	7	5	0	0	8
Exhaust Port Temperature 2	65186	254	162	0	7	1	1000	0	8
Exhaust Port Temperature 5	65183	254	159	0	7	1	1000	0	8
Turbocharger Information 1	65179	254	155	0	7	1	1000	0	8
MTU Air Start Pressure	65502	255	222	0	6	5	0	0	8
MTU Eng Contr 1	65360	255	80	0	3	4	100	100	8
Engine Configuration 1	65251	254	227	0	6	4	2000	100	39
Exhaust Temperature	65031	254	7	0	6	1	500	0	8
Trip Time Information 2	65200	254	176	0	7	5	0	0	20
Engine Fluid Level/Pressure #1	65263	254	239	0	6	1	500	0	8
MTU Eng Fluid Level Pressure 1	65363	255	83	0	6	1	500	0	8
MTU Eng Contr 4 - Addi Inform	65284	255	4	0	6	1	100	0	8
Fuel Consumption (Liquid)	65257	254	233	0	6	5	0	0	8
Engine Information	65170	254	146	0	7	1	100	0	8
Shutdown	65252	254	228	0	6	1	1000	0	8
MTU Eng Contr 2 - Speed Dem FB	65361	255	81	0	6	1	100	0	8
Electronic Engine Controller 2	61443	240	3	0	3	1	50	0	8
Exhaust Port Temperature 1	65187	254	163	0	7	1	1000	0	8
Fuel Economy (Liquid)	65266	254	242	0	6	1	100	0	8
Engine Temperature 2	65188	254	164	0	6	1	1000	0	8
Electronic Engine Controller 3	65247	254	223	0	6	1	250	0	8
Inlet/Exhaust Conditions 1	65270	254	246	0	6	1	500	0	8
Inlet/Exhaust Conditions 2	64976	253	208	0	6	1	500	0	8
Vehicle Electrical Power1	65271	254	247	0	6	1	1000	0	8
Electronic Engine Controller 4	65214	254	190	0	7	5	0	0	8
Engine Fuel / Lube Systems	65130	254	106	0	6	1	500	0	8
MTU Eng Contr 5 - Stat Signals	65367	255	87	0	6	1	500	0	8
MTU Engine Temperature 1	65364	255	84	0	6	1	1000	0	8
MTU Engine Temperature 2	65373	255	93	0	6	1	1000	0	8
Turbocharger	65245	254	221	0	6	1	1000	0	8
Exhaust Port Temperature 4	65184	254	160	0	7	1	1000	0	8
MTU Pressure Limits 1	65395	255	115	0	6	5	0	0	8
MTU Pressure Limits 2	65396	255	116	0	6	5	0	0	8
MTU Pressure Limits 3	65397	255	117	0	6	5	0	0	8
MTU Pressure Limits 4	65439	255	159	0	6	5	0	0	8
MTU Turbocharger 1 and 2 Speed Limits	65398	255	118	0	6	5	0	0	8
MTU Turbocharger 3 and 4 Speed Limits	65399	255	119	0	6	5	0	0	8
MTU Temperature Limits 1	65402	255	122	0	6	5	0	0	8
MTU Temperature Limits 2	65403	255	123	0	6	5	0	0	8
MTU-ETC-Information	65404	255	124	0	6	1	200	0	8
MTU Fuel Consumption Info	65405	255	125	0	6	1	500	0	8
MTU Splash Oil Temperature Limits 1	65406	255	126	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 2	65407	255	127	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 3	65408	255	128	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 4	65409	255	129	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 5	65410	255	130	0	6	5	0	0	8
Main Bearing Temperature 1	65182	254	158	0	6	1	1000	0	8
Main Bearing Temperature 2	65181	254	157	0	6	1	1000	0	8
Main Bearing Temperature 3	65180	254	156	0	6	1	1000	0	8
Ambient Conditions	65269	254	245	0	6	1	1000	0	8
MTU Coolant Temperature (EMU) Limits	65413	255	133	0	6	5	0	0	8
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	255	132	0	6	5	0	0	8
Engine Auxiliary Coolant	65172	254	148	0	6	4	500	100	8
MTU Speed Limits	65415	255	135	0	6	5	0	0	8
MTU Refill Pump	65418	255	138	0	6	1	500	0	8
Time/Date	65254	254	230	0	6	5	0	0	8



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PGN-Name	PGN	PDUFormat	PDUSpecific	DataPage	DefaultPriority	TransferMode	TxRepetitionRate	SendOnChangeTime	DLC
MTU System DM1 Producer Extension	65483	255	203	0	6	4	1000	100	8
MTU User Information	65421	255	141	0	6	1	100	0	8
MTU Splash Oil Temperature 3	65371	255	91	0	6	1	1000	0	8
Engine Average Information	64851	253	83	0	5	1	500	0	8
MTU Splash Oil Temperature 4	65370	255	90	0	6	1	1000	0	8
MTU Exhaust Temperature Limits	65411	255	131	0	6	5	0	0	8
MTU Splash Oil Temperature 5	65369	255	89	0	6	1	1000	0	8
Aftertreatment 1 Diesel Exhaust Fluid Tank Volume	65110	254	86	0	6	1	1000	0	8
MTU Several Limits	65442	255	162	0	6	1	2000	0	8
Aftertreatment 1 Diesel Exhaust Fluid Information 1	64923	253	155	0	6	1	500	0	8
Aftertreatment 1 SCR Exhaust Gas Temperature 1	64830	253	62	0	6	1	500	0	8
Aftertreatment 2 SCR Exhaust Gas Temperature 1	64824	253	56	0	6	1	500	0	8
MTU SCR Limits	65443	255	163	0	6	1	2000	0	8
Aftertreatment 1 SCR Dosing System Information 1	61475	240	35	0	6	1	500	0	8
Aftertreatment 1 SCR Exhaust Gas Pressures	64831	253	63	0	6	1	500	0	8
EES / Electrical Load	65171	254	147	0	6	1	500	0	8
AT / Alternator Temperature	65191	254	167	0	7	1	1000	0	8
Alarm Groups	65226	254	202	0	6	1	500	0	8
T-Exhaust Combined Limits	65400	255	120	0	6	5	0	0	8
T-Winding Limits	65419	255	139	0	6	5	0	0	24
MTU Binary Signals	65420	255	140	0	6	1	500	0	8
MTU Generator Temperatures 1	65445	255	165	0	6	1	1000	0	8
Dash Display 2	64671	252	159	0	6	1	1000	0	8

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J1939 PGN Rx Signals

PGN-Name	PGN	PDUFormat	PDUSpecific	DataPage	DefaultPriority	TransferMode	TxRepetitionRate	SendOnChangeTime	DLC
MTU System Operating PDU1 prop A1	61184	239	0	0	6	4	2000	100	8

J1939 SPN Tx Signals

PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
Engine Hours, Revolutions	65253	Engine Total Hours of Operation	247	32	1	1	58
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 9 Temperature	1145	16	1	1	68
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 10 Temperature	1146	16	3	1	68
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 11 Temperature	1147	16	5	1	68
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 12 Temperature	1148	16	7	1	68
Dash Display	65276	Engine Fuel Filter Differential Pressure	95	8	3	1	53
Dash Display	65276	Engine Oil Filter Differential Pressure	99	8	4	1	54
Engine Electronic Controller #1	61444	Engine Speed	190	16	4	1	76
Engine Temperature 1	65262	Engine Coolant Temperature	110	8	1	1	67
Engine Temperature 1	65262	Engine Fuel 1 Temperature 1	174	8	2	1	67
Engine Temperature 1	65262	Engine Oil Temperature 1	175	16	3	1	68
Engine Temperature 1	65262	Engine Turbocharger Oil Temperature	176	16	5	1	68
Fuel Information (Liquid)	65203	Trip Average Fuel Rate	1029	16	5	1	42
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 5 Temperature	1141	16	1	1	68
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 6 Temperature	1142	16	3	1	68
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 7 Temperature	1143	16	5	1	68
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 8 Temperature	1144	16	7	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 17 Temperature	1153	16	1	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 18 Temperature	1154	16	3	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 19 Temperature	1155	16	5	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 20 Temperature	1156	16	7	1	68
Turbocharger Information 1	65179	Engine Turbocharger Lube Oil Pressure 2	1168	8	1	1	47
Turbocharger Information 1	65179	Engine Turbocharger 2 Speed	1169	16	2	1	77
Turbocharger Information 1	65179	Engine Turbocharger 3 Speed	1170	16	4	1	77
Turbocharger Information 1	65179	Engine Turbocharger 4 Speed	1171	16	6	1	77
MTU Air Start Pressure	65502	MTU Engine Air Start Pressure (Extended Range)	520274	16	1	1	106
MTU Eng Contr 1	65360	MTU Engine Speed redundant	520382	16	1	1	76
MTU Eng Contr 1	65360	Engine Turning Status	520200	2	3	1	87
MTU Eng Contr 1	65360	Engine Safety and Protection Override Status	520202	2	3	3	87
MTU Eng Contr 1	65360	Engine Local Operation Status	520203	2	3	5	87
MTU Eng Contr 1	65360	MTU Engine Running State	520255	2	3	7	87
MTU Eng Contr 1	65360	Engine Cylinder Cutoff	520252	2	5	1	87
MTU Eng Contr 1	65360	Engine Turbocharger 2 Status	520237	2	5	3	87
MTU Eng Contr 1	65360	Engine Turbocharger 3 Status	520242	2	8	1	87
MTU Eng Contr 1	65360	Engine Turbocharger 4 Status	520243	2	8	3	87
Engine Configuration 1	65251	Engine Speed At Idle, Point 1	188	16	1	1	76
Exhaust Temperature	65031	Engine Exhaust Manifold Bank 2 Temperature 1	2433	16	1	1	68
Exhaust Temperature	65031	Engine Exhaust Manifold Bank 1 Temperature 1	2434	16	3	1	68
Trip Time Information 2	65200	Trip Engine Running Time	1036	32	9	1	58
Engine Fluid Level/Pressure #1	65263	Engine Fuel Delivery Pressure	94	8	1	1	47
Engine Fluid Level/Pressure #1	65263	Engine Oil Pressure	100	8	4	1	47
Engine Fluid Level/Pressure #1	65263	Engine Crankcase Pressure 1	101	16	5	1	52
Engine Fluid Level/Pressure #1	65263	Engine Coolant Pressure 1	109	8	7	1	53
MTU Eng Fluid Level Pressure 1	65363	MTU Engine Oil Pressure redundant	520292	8	1	1	47
MTU Eng Fluid Level Pressure 1	65363	MTU Engine Fuel Delivery Pressure behind Filter	520286	8	2	1	47
MTU Eng Fluid Level Pressure 1	65363	MTU Engine Fuel Pre-Filter Differential Pressure	520287	8	3	1	53
MTU Eng Contr 4 - Addtl Inform	65284	MTU Engine Controller Error Codes	520256	16	1	1	12
Fuel Consumption (Liquid)	65257	Engine Trip Fuel	182	32	1	1	20
Fuel Consumption (Liquid)	65257	Engine Total Fuel Used	250	32	5	1	20
Engine Information	65170	Engine Oil Filter Intake Pressure	1208	8	1	1	47
Shutdown	65252	Engine Protection System has Shutdown Engine	1110	2	5	1	87



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PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
Shutdown	65252	Engine Protection System Approaching Shutdown	1109	2	5	3	87
Shutdown	65252	Engine Overspeed Test	2812	2	7	7	87
MTU Eng Contr 2 - Speed Dem FB	65361	Engine Demanded Operating Speed	520707	16	1	1	76
Electronic Engine Controller 2	61443	Engine Percent Load At Current Speed	92	8	3	1	46
Electronic Engine Controller 2	61443	Actual Maximum Available Engine - Percent Torque	3357	8	7	1	44
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 1 Temperature	1137	16	1	1	68
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 2 Temperature	1138	16	3	1	68
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 3 Temperature	1139	16	5	1	68
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 4 Temperature	1140	16	7	1	68
Fuel Economy (Liquid)	65266	Engine Fuel Rate	183	16	1	1	42
Engine Temperature 2	65188	Engine ECU Temperature	1136	16	3	1	68
Electronic Engine Controller 3	65247	Engine's Desired Operating Speed	515	16	2	1	76
Inlet/Exhaust Conditions 1	65270	Engine Intake Manifold 1 Temperature	105	8	3	1	67
Inlet/Exhaust Conditions 2	64976	Engine Intake Manifold #1 Absolute Pressure	3563	8	5	1	53
Vehicle Electrical Power1	65271	Battery Potential / Power Input 1	168	16	5	1	80
Vehicle Electrical Power1	65271	Key Switch Battery Potential	158	16	7	1	80
Electronic Engine Controller 4	65214	Engine Rated Speed	189	16	3	1	76
Engine Fuel / Lube Systems	65130	Engine Oil Priming Pump Switch	3550	2	6	1	87
MTU Eng Contr 5 - Stat Signals	65367	Engine Status of Startprocedure	520241	32	3	1	245
MTU Eng Contr 5 - Stat Signals	65367	MTU Start Interlock	520266	2	7	1	87
MTU Engine Temperature 1	65364	MTU Engine Coolant Temperature redundant	520302	8	1	1	67
MTU Engine Temperature 1	65364	Engine Charge Air Sequential Control Valve Temperature	520254	16	2	1	68
MTU Engine Temperature 1	65364	Engine Main Bearing Temperature Average	520407	16	4	1	68
MTU Engine Temperature 2	65373	Engine Exhaust Gas Temperature - Left 2	520393	16	1	1	68
MTU Engine Temperature 2	65373	Engine Exhaust Gas Temperature - Right 2	520394	16	3	1	68
Turbocharger	65245	Engine Turbocharger Lube Oil Pressure 1	104	8	1	1	47
Turbocharger	65245	Engine Turbocharger 1 Speed	103	16	2	1	77
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 13 Temperature	1149	16	1	1	68
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 14 Temperature	1150	16	3	1	68
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 15 Temperature	1151	16	5	1	68
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 16 Temperature	1152	16	7	1	68
MTU Pressure Limits 1	65395	LM LO P-Lube Oil	520500	8	1	1	47
MTU Pressure Limits 1	65395	LM LOLO P-Lube Oil	520501	8	2	1	47
MTU Pressure Limits 1	65395	LM HI P-Crankcase	520502	16	3	1	52
MTU Pressure Limits 1	65395	LM HIHI P-Crankcase	520503	16	5	1	52
MTU Pressure Limits 1	65395	LM LO P-Coolant	520504	8	7	1	53
MTU Pressure Limits 1	65395	LM LOLO P-Coolant	520505	8	8	1	53
MTU Pressure Limits 2	65396	LM LO P-Fuel	520506	8	1	1	47
MTU Pressure Limits 2	65396	LM LOLO P-Fuel	520507	8	2	1	47
MTU Pressure Limits 2	65396	LM HI P-Charge Air	520508	8	3	1	53
MTU Pressure Limits 2	65396	LM HI P-Lube Oil Filter Diff.	520537	8	5	1	54
MTU Pressure Limits 2	65396	LM HI P-Fuel Filter Diff.	520538	8	6	1	54
MTU Pressure Limits 2	65396	LM LO Limit Engine Turbocharger Lube Oil Pressure 1	520470	8	8	1	47
MTU Pressure Limits 3	65397	LM LO P-Start Air	520540	16	1	1	53
MTU Pressure Limits 3	65397	LM LOLO P-Start Air Start Term	520541	16	3	1	47
MTU Pressure Limits 3	65397	LM LO P-Oil Refill Pump	520542	8	5	1	47
MTU Pressure Limits 3	65397	LM LOLO Limit Engine Turbocharger Lube Oil Pressure 1	520471	8	6	1	47
MTU Pressure Limits 3	65397	LM LO Limit Engine Turbocharger Lube Oil Pressure 2	520472	8	7	1	47
MTU Pressure Limits 3	65397	LM LOLO Limit Engine Turbocharger Lube Oil Pressure 2	520473	8	8	1	47
MTU Pressure Limits 4	65439	LM HIHI Lube Oil Filter Diff Pressure	521580	8	3	1	47
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HI ETC 1 Speed	520510	16	1	1	77
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HIHI ETC 1 Speed	520511	16	3	1	77
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HI ETC 2 Speed	520512	16	5	1	77
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HIHI ETC 2 Speed	520513	16	7	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HI ETC 3 Speed	520514	16	1	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HIHI ETC 3 Speed	520515	16	3	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HI ETC 4 Speed	520516	16	5	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HIHI ETC 4 Speed	520517	16	7	1	77
MTU Temperature Limits 1	65402	LM HI T-Coolant (ECU)	520518	8	1	1	67
MTU Temperature Limits 1	65402	LM HIHI T-Coolant (ECU)	520519	8	2	1	67



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PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
MTU Temperature Limits 1	65402	LM HI T-Charge Air	520520	8	3	1	67
MTU Temperature Limits 1	65402	LM HIHI T-Charge Air	520521	8	4	1	67
MTU Temperature Limits 1	65402	LM HI T-Lube Oil	520522	16	5	1	68
MTU Temperature Limits 1	65402	LM HIHI T-Lube Oil	520523	16	7	1	68
MTU Temperature Limits 2	65403	LM HI T-Charge Air Seq Ctrl Valve	520532	16	1	1	68
MTU Temperature Limits 2	65403	LM HI T-Fuel	520534	8	3	1	67
MTU Temperature Limits 2	65403	LM HIHI T-Fuel	520535	8	4	1	67
MTU Temperature Limits 2	65403	UL T-ECU	520536	16	5	1	68
MTU Temperature Limits 2	65403	LM HIHI T-Charge Air Seq Ctrl Valve	520533	16	7	1	68
MTU-ETC-Information	65404	ETC 1 Running	520560	2	2	1	87
MTU-ETC-Information	65404	ETC 2 Running	520561	2	2	3	87
MTU-ETC-Information	65404	ETC 3 Running	520562	2	2	5	87
MTU-ETC-Information	65404	ETC 4 Running	520563	2	2	7	87
MTU Fuel Consumption Info	65405	Max. Scale Fuel Consumpt./h	520564	16	1	1	42
MTU Fuel Consumption Info	65405	Actual Fuel Cons. %/l	520565	8	3	1	44
MTU Fuel Consumption Info	65405	Mean Trip Fuel Cons. %/l	520566	8	4	1	44
MTU Splash Oil Temperature Limits 1	65406	LM HI T-Splash Oil B1	520362	16	1	1	68
MTU Splash Oil Temperature Limits 1	65406	LM HIHI T-Splash Oil B1	520363	16	3	1	68
MTU Splash Oil Temperature Limits 1	65406	LM HI T-Splash Oil B2	520364	16	5	1	68
MTU Splash Oil Temperature Limits 1	65406	LM HIHI T-Splash Oil B2	520365	16	7	1	68
MTU Splash Oil Temperature Limits 2	65407	LM HI T-Splash Oil B3	520366	16	1	1	68
MTU Splash Oil Temperature Limits 2	65407	LM HIHI T-Splash Oil B3	520367	16	3	1	68
MTU Splash Oil Temperature Limits 2	65407	LM HI T-Splash Oil B4	520368	16	5	1	68
MTU Splash Oil Temperature Limits 2	65407	LM HIHI T-Splash Oil B4	520369	16	7	1	68
MTU Splash Oil Temperature Limits 3	65408	LM HI T-Splash Oil B5	520370	16	1	1	68
MTU Splash Oil Temperature Limits 3	65408	LM HIHI T-Splash Oil B5	520371	16	3	1	68
MTU Splash Oil Temperature Limits 3	65408	LM HI T-Splash Oil B6	520372	16	5	1	68
MTU Splash Oil Temperature Limits 3	65408	LM HIHI T-Splash Oil B6	520373	16	7	1	68
MTU Splash Oil Temperature Limits 4	65409	LM HI T-Splash Oil B7	520374	16	1	1	68
MTU Splash Oil Temperature Limits 4	65409	LM HIHI T-Splash Oil B7	520375	16	3	1	68
MTU Splash Oil Temperature Limits 4	65409	LM HI T-Splash Oil B8	520376	16	5	1	68
MTU Splash Oil Temperature Limits 4	65409	LM HIHI T-Splash Oil B8	520377	16	7	1	68
MTU Splash Oil Temperature Limits 5	65410	LM HI T-Splash Oil B9	520378	16	1	1	68
MTU Splash Oil Temperature Limits 5	65410	LM HIHI T-Splash Oil B9	520379	16	3	1	68
MTU Splash Oil Temperature Limits 5	65410	LM HI T-Splash Oil B10	520380	16	5	1	68
MTU Splash Oil Temperature Limits 5	65410	LM HIHI T-Splash Oil B10	520381	16	7	1	68
Main Bearing Temperature 1	65182	Engine Main Bearing 1 Temperature	1157	16	1	1	68
Main Bearing Temperature 1	65182	Engine Main Bearing 2 Temperature	1158	16	3	1	68
Main Bearing Temperature 1	65182	Engine Main Bearing 3 Temperature	1159	16	5	1	68
Main Bearing Temperature 1	65182	Engine Main Bearing 4 Temperature	1160	16	7	1	68
Main Bearing Temperature 2	65181	Engine Main Bearing 5 Temperature	1161	16	1	1	68
Main Bearing Temperature 2	65181	Engine Main Bearing 6 Temperature	1162	16	3	1	68
Main Bearing Temperature 2	65181	Engine Main Bearing 7 Temperature	1163	16	5	1	68
Main Bearing Temperature 2	65181	Engine Main Bearing 8 Temperature	1164	16	7	1	68
Main Bearing Temperature 3	65180	Engine Main Bearing 9 Temperature	1165	16	1	1	68
Main Bearing Temperature 3	65180	Engine Main Bearing 10 Temperature	1166	16	3	1	68
Main Bearing Temperature 3	65180	Engine Main Bearing 11 Temperature	1167	16	5	1	68
Ambient Conditions	65269	Engine Intake 1 Air Temperature	172	8	6	1	67
MTU Coolant Temperature (EMU) Limits	65413	LM HI T-Coolant (EMU)	520352	8	1	1	67
MTU Coolant Temperature (EMU) Limits	65413	LM HIHI T-Coolant (EMU)	520353	8	2	1	67
MTU Coolant Temperature (EMU) Limits	65413	LM HI Engine Air Inlet Temperature	520475	8	3	1	67
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HI T-Bearing Mean	520358	16	1	1	68
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HI T-Bearing	520359	16	3	1	68
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HIHI T-Bearing	520360	16	5	1	68
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM LO P-Raw Water	520361	8	7	1	53
MTU Bearing Temperature and Raw-Water Pressure Limits		LM HIHI P-Crankcase	520474	8	8	1	53
Engine Auxiliary Coolant	65172	Sea Water Pump Outlet Pressure	2435	8	3	1	53
MTU Speed Limits	65415	LM HIHI Engine Speed	520387	16	1	1	76
MTU Speed Limits	65415	UL Engine Speed (EMU)	520725	16	3	1	76
MTU Refill Pump	65418	P-Oil Refill Pump	520567	8	1	1	47
MTU Refill Pump	65418	P-Fuel Prefilter	521595	8	3	1	54
MTU Refill Pump	65418	P-DEF Filter Out at PDM	521438	16	4	1	85



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-22Aenderungs-Nr./Revision Notice No.
PR033847Bearbeitungsstatus/Lifecycle
Released

PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
Time/Date	65254	Seconds	959	8	1	1	63
Time/Date	65254	Minutes	960	8	2	1	61
Time/Date	65254	Hours	961	8	3	1	3
Time/Date	65254	Month	963	8	4	1	62
Time/Date	65254	Day	962	8	5	1	60
Time/Date	65254	Year	964	8	6	1	66
MTU System DM1 Producer Extension	65483	MTU DM1 Horn Status	520815	2	1	1	87
MTU User Information	65421	User Information	520714	16	1	1	276
MTU Splash Oil Temperature 3	65371	MTU Splash Oil Temperature Mean	520467	16	1	1	68
MTU Splash Oil Temperature 3	65371	Splash Oil Temperature 11	520395	16	5	1	68
MTU Splash Oil Temperature 3	65371	Splash Oil Temperature 12	520396	16	7	1	68
Engine Average Information	64851	Engine Exhaust Temperature Average	4151	16	1	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 13	520397	16	1	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 14	520398	16	3	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 15	520399	16	5	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 16	520400	16	7	1	68
MTU Exhaust Temperature Limits	65411	LM HI T-Exhaust	520354	16	1	1	68
MTU Exhaust Temperature Limits	65411	LM LO T-Exhaust	520355	16	3	1	68
MTU Exhaust Temperature Limits	65411	LM HIHI T-Exhaust	520356	16	5	1	68
MTU Exhaust Temperature Limits	65411	LM LOLO T-Exhaust	520357	16	7	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 17	520401	16	1	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 18	520402	16	3	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 19	520403	16	5	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 20	520404	16	7	1	68
Aftertreatment 1 Diesel Exhaust Fluid Tank Volume	65110	Aftertreatment 1 Diesel Exhaust Fluid Tank Volume	1761	8	1	1	44
MTU Several Limits	65442	LM LO P-DEF Filter Out at PDM	521586	16	1	1	85
MTU Several Limits	65442	LM HI T-DEF at PDM	521587	8	3	1	67
MTU Several Limits	65442	LM LO T-DEF at PDM	521588	8	4	1	67
MTU Several Limits	65442	LM LO DEF Tank Level	521589	8	5	1	44
MTU Several Limits	65442	LM LOLO DEF Tank Level	521590	8	6	1	44
MTU Several Limits	65442	LM HI P-Fuel Prefilter	521591	8	7	1	54
MTU Several Limits	65442	LM HIHI P-Fuel Prefilter	521592	8	8	1	54
Aftertreatment 1 Diesel Exhaust Fluid Information 1	64923	Aftertreatment 1 Diesel Exhaust Fluid Temperature 2	3515	8	2	1	67
Aftertreatment 1 SCR Exhaust Gas Temperature 1	64830	Aftertreatment 1 SCR Intake Temperature	4360	16	1	1	68
Aftertreatment 1 SCR Exhaust Gas Temperature 1	64830	Aftertreatment 1 SCR Outlet Temperature	4363	16	4	1	68
Aftertreatment 2 SCR Exhaust Gas Temperature 1	64824	Aftertreatment 2 SCR Intake Temperature	4413	16	1	1	68
MTU SCR Limits	65443	LM HI T-Exhaust SCR	521582	16	1	1	68
MTU SCR Limits	65443	LM HIHI T-Exhaust SCR	521583	16	3	1	68
MTU SCR Limits	65443	LM HI P-Diff Exhaust SCR 1	521584	16	5	1	85
MTU SCR Limits	65443	LM HIHI P-Diff Exhaust SCR 1	521585	16	7	1	85
Aftertreatment 1 SCR Dosing System Information 1	61475	Aftertreatment 1 SCR System 1 State	4332	4	3	1	89
Aftertreatment 1 SCR Exhaust Gas Pressures	64831	Aftertreatment 1 SCR Differential Pressure	4358	16	1	1	85
EES / Electrical Load	65171	Electrical Load	1204	16	1	1	55
AT / Alternator Temperature	65191	Engine Alternator Bearing 1 Temperature	1122	8	1	1	67
AT / Alternator Temperature	65191	Engine Alternator Bearing 2 Temperature	1123	8	2	1	67
AT / Alternator Temperature	65191	Engine Alternator Winding 1 Temperature	1124	8	3	1	67
AT / Alternator Temperature	65191	Engine Alternator Winding 2 Temperature	1125	8	4	1	67
AT / Alternator Temperature	65191	Engine Alternator Winding 3 Temperature	1126	8	5	1	67
Alarm Groups	65226	Red Stop Lamp	623	32	1	1	245
Alarm Groups	65226	Amber Warning Lamp	624	32	5	1	245
T-Exhaust Combined Limits	65400	LM HI T-Exhaust Combined A	520524	16	1	1	68
T-Exhaust Combined Limits	65400	LM HIHI T-Exhaust Combined A	520525	16	3	1	68
T-Exhaust Combined Limits	65400	LM HI T-Exhaust Combined B	520526	16	5	1	68
T-Exhaust Combined Limits	65400	LM HIHI T-Exhaust Combined B	520527	16	7	1	68
T-Winding Limits	65419	LM HIHI T-Winding L1	520726	8	1	1	67
T-Winding Limits	65419	LM HI T-Winding L1	520727	8	2	1	67
T-Winding Limits	65419	LM HIHI T-Winding L3	520728	8	3	1	67
T-Winding Limits	65419	LM HI T-Winding L3	520729	8	4	1	67
T-Winding Limits	65419	LM HIHI T-Winding L2	520730	8	5	1	67
T-Winding Limits	65419	LM HI T-Winding L2	520731	8	6	1	67
T-Winding Limits	65419	LM LOLO P-Lube Oil (EMU)	520723	8	7	1	47
T-Winding Limits	65419	LM LO P-Lube Oil (EMU)	520724	8	8	1	47
MTU Generator Temperatures 1	65445	T-Coolant Gen Out	521810	8	3	1	68



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Aenderungsbeschreibung/Description of Revision war kZ		Kommt vor/Frequency
Buchst./Rev.Ltr. -22	Aenderungs-Nr./Revision Notice No. PR033847	Bearbeitungsstatus/Lifecycle Released

PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
MTU Generator Temperatures 1	65445	T-Air Inlet Gen	521839	8	1	1	67
MTU Generator Temperatures 1	65445	T-Air Outlet Gen	521840	8	2	1	67
MTU Binary Signals	65420	Ready For Start Stop FMCS	520718	2	1	7	87
MTU Binary Signals	65420	Engine Stop Active	520719	2	2	1	87
MTU Binary Signals	65420	Start Active	520720	2	2	3	87
MTU Generator Temperatures 1	65445	T-Coolant Gen In	521811	8	4	1	68
T-Winding Limits	65419	LM HI T-Bearing GenDE	521816	8	21	1	68
T-Winding Limits	65419	LM HIHI T-Bearing GenDE	521817	8	22	1	68
T-Winding Limits	65419	LM HI T-Bearing GenNDE	521818	8	23	1	68
T-Winding Limits	65419	LM HIHI T-Bearing GenNDE	521819	8	24	1	68
T-Winding Limits	65419	LM HI T-AirInlet Gen	521820	8	17	1	68
T-Winding Limits	65419	LM HIHI T-AirInlet Gen	521821	8	18	1	68
T-Winding Limits	65419	LM HI T-AirOutlet Gen	521822	8	19	1	68
T-Winding Limits	65419	LM HIHI T-AirOutlet Gen	521823	8	20	1	68
MTU Binary Signals	65420	FIFI Mode Active	521824	2	3	1	87
MTU Binary Signals	65420	Cmd Generator Shutdown	521825	2	3	3	87
T-Winding Limits	65419	LM HI T-Coolant Gen Out	521812	8	13	1	68
T-Winding Limits	65419	LM HI T-Coolant Gen In	521813	8	14	1	68
T-Winding Limits	65419	LM HIHI T-Coolant Gen Out	521814	8	15	1	68
T-Winding Limits	65419	LM HIHI T-Coolant Gen In	521815	8	16	1	68
Dash Display 2	64671	Engine Oil Filter Differential Pressure (Extended)	6321	16	1	1	52
MTU Eng Contr 4 - Addi Inform	65284	Engine Load Reserve	521826	8	8	1	44
Engine Electronic Controller #1	61444	Injection Quantity 0-120	512	8	2	1	44
MTU Eng Fluid Level Pressure 1	65363	P-Exhaust Back	521828	16	6	1	52
T-Winding Limits	65419	LM HI P-Exhaust Back	521829	16	9	1	52
T-Winding Limits	65419	LM HIHI P-Exhaust Back	521830	16	11	1	52

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Aenderungs-Nr./Revision Notice No.
PR033847

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J1939 SPN Rx Signals

PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
MTU System Operating PDU1 prop A1	61184	Engine Start Command	520192	2	1	1	87
MTU System Operating PDU1 prop A1	61184	Engine Stop Command	520193	2	1	3	87
MTU System Operating PDU1 prop A1	61184	Engine Safety and Protection Override Command	520194	2	1	5	87
MTU System Operating PDU1 prop A1	61184	MTU Stop Priming Command	520213	2	1	7	87
MTU System Operating PDU1 prop A1	61184	MTU DM1 Global DTC Acknowledge Command	520265	2	2	1	87
MTU System Operating PDU1 prop A1	61184	MTU DM1 Horn Command	520816	2	2	3	87
MTU System Operating PDU1 prop A1	61184	MTU RESET Trip Command	988	2	2	7	87
MTU System Operating PDU1 prop A1	61184	Engine Turn Command	520196	2	3	1	87
MTU System Operating PDU1 prop A1	61184	Engine Overspeed Test Command	520197	2	3	3	87
MTU System Operating PDU1 prop A1	61184	Harbour Test Mode Command	520199	2	3	5	87

J1939 Slot List NMEA2000 Data Format

SLOT Identifier	SLOT Name	SLOT Type	Scaling	Range	Offset	Length
1	SAEpr11	Pressure	5 kPa/bit	0 to 1,250 kPa	0	1 byte
2	SAEpr13	Pressure	8 kPa/bit	0 to 2,000 kPa	0	1 byte
3	SAEtm11	Time	1 hr/bit	0 to 250 hr	0	1 byte
4	SAEtm10	Time	1 hr/bit	-125 to 125 hr	-125 hr	1 byte
5	SAEtm12	Time	1 hr/bit	-32,127 to 32,128 hr	-32,127 hr	2 bytes
6	SAEtm06	Time	1 s/bit	0 to 4,211,081,215 s	0	4 bytes
7	SAEad01	Angle/Direction	10 ⁻⁷ deg/bit	-210 to 211.1081215 deg	-210 deg	4 bytes
8	SAEad04	Angle/Direction	1/128 deg/bit	-200 to 301.99 deg	-200 deg	2 bytes
9	SAEad05	Angle/Direction	1/128 deg/bit	0 to 501.99 deg	0	2 bytes
10	SAEad02	Angle/Direction	0.1 sec/bit	-3276.8 to 3148.7 sec	-3,276.8 sec	2 bytes
11	SAEad06	Angle/Direction	1 deg/bit	-125 to 125 deg	-125 deg	1 byte
12	SAEct04	Count	1 count/bit	0 to 64,255	0	2 bytes
13	SAEds03	Distance	0.1 mm/bit	-3,200 to 3,225.5 mm	-3,200 mm	2 bytes
14	SAEds04	Distance	0.1 mm/bit	0 to 6,425.5 mm (0 to 6.4255 m)	0	2 bytes
15	SAEds08	Distance	1 m/bit	-125 to 125 m	-125 m	1 byte
16	SAEel01	Economy, liquid	1/512 km/L per bit	0 to 125.498046875 km/L	0	2 bytes
17	SAEeg01	Economy, gaseous	1/512 km/kg per bit	0 to 125.498046875 km/kg	0	2 bytes
18	SAEfg01	Flow rate, gaseous	0.05 kg/h per bit	0 to 3212.75 kg/h	0	2 bytes
19	SAEf01	Flow rate, volumetric	0.1 m ³ /h per bit	0 to 6425.5 m ³ /h	0	2 bytes
20	SAEfu01	Fuel Used, liquid	0.5 L/bit	0 to 2,105,540,607.5 L	0	4 bytes
21	SAEfug01	Fuel Used, gaseous	0.5 kg/bit	0 to 2,105,540,607.5 kg	0	4 bytes
22	SAEgg01	Governor gain	1/1280 %/rpm per bit	0 to 50.2 %/rpm	0	2 bytes
23	SAEgr01	Gear Ratio	0.01/bit	0 to 642.55	0	2 bytes
24	SAEgv01	Gear Value	1 gear value/bit	-125 to 125	-125	1 byte
25	SAEid01	Identifier, component/software	1 ID/bit	0 to 250 ID	0	1 byte
26	SAEkv01	Kinematic viscosity	1 mm ² /s per bit	0 to 250 mm ² /s	0	1 byte
27	SAEma11	Mass	0.5 kg/bit	0 to 32,127.5 kg	0	2 bytes
28	SAEma02	Mass	2 kg/bit	0 to 128,510 kg	0	2 bytes
29	SAEma03	Mass	10 kg/bit	0 to 642,550 kg	0	2 bytes
30	SAEpc01	Percent, position/level	0.0025 %/bit	0 to 160.6375 %	0	2 bytes
31	SAEr02	Ratio	0.1/bit	0 to 25.0	0	1 byte
32	SAEr01	Ratio	0.001/bit	0 to 64,255	0	2 bytes
33	SAEre01	Record	1 record/bit	0 to 250 records	0	1 byte
34	SAEr01	Revolutions	1000 r/bit	0 to 4,211,081,215,000 r	0	4 bytes
35	SAEsa01	Source Address	1 source address/bit	0 to 255	0	1 byte
36	SAEds10	Distance	0.125 km/bit	0 to 526,385,151.875 km	0	4 bytes
37	SAEds11	Distance	5 km/bit	-160,635 to 160,640 km	-160,635 km	2 bytes
38	SAEds09	Distance	5 m/bit	0 to 21,055,406 km	0	4 bytes
39	SAEds06	Distance	0.125 m/bit	-2500 to 5531.875 m	-2500 m	2 bytes
40	SAEec02	Electrical Current	1 A/bit	-125 to 125 A	-125 A	1 byte
41	SAEec03	Electrical Current	1 A/bit	0 to 250 A	0	1 byte
42	SAEfl01	Flow rate, liquid	0.05 L/h per bit	0 to 3,212.75 L/h	0	2 bytes
43	SAEfr01	Force	5 N/bit	0 to 321,275 N	0	2 bytes
44	SAEpc03	Percent, position/level	0.4 %/bit	0 to 100 %	0	1 byte
45	SAEpc05	Percent, position/level	1 %/bit	-125 to 125 %	-125 %	1 byte
46	SAEpc06	Percent, position/level	1 %/bit	0 to 250 %	0	1 byte



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